Documentation Structure

Layout

- Cover Page (Front Matter, Statement of Submission, Acknowledgement, Abstract)
- Table of Contents
- List of Tables
- List of Figures
- **Chapter 1** - Introduction (Subtopics are Background, Problem Statement, Objectives, Scope, Methodology, Technologies)
- **Chapter 2** - Literature Review (To compare with Google Classroom, Microsoft Teams, Moodle Notifications, LMS communication modules, Email campaign systems, Slack/Discord notification concepts, AI spam filtering systems)
- **Chapter 3** - System Analysis (Includes Functional Requirements, Non-Functional Requirements, Use Case Diagrams, User Roles, Domain Model Diagram, DFDs)
- **Chapter 4** - System Design (Sections to be included are System Architecture, Database Design, ERDs, Interface Design, Component Design, Class Diagram, Object Diagram, Sequence Diagram. Additionally, sections to be noted/included are Distributed System Architecture, Service Communication Architecture, Frontend Architecture, Backend-core Architecture, Realtime Architecture, AI Architecture, Automation Rule Engine Design, Security Architecture)
- **Chapter 5** - Implementation (Sections to be included are Development Tools and Environment, Modules/Component Implemented. Additionally, Frontend Implementation, Backend-Core Implementation, Realtime Service Implementation, AI Service Implementation, Database Implementation, Authentication & Authorization, Rule Engine Implementation, Campaign Management Implementation, AI Classification Implementation, Realtime Messaging Implementation)
- **Chapter 6** - AI/ML Methodology (Includes dataset selection, preprocessing, feature extraction, training process, evaluation metrics, inference workflow)
- **Chapter 7** - Deployment Architecture (It'd include deployment topology, hosting strategy, cloud architecture, environment separation, CI/CD overview)
- **Chapter 8** - System Testing (Sections are Testing Techniques Used, Test Plans, Test Cases, Results and Bug Reports. Additionally are Unit Testing, API Testing, Realtime Communication, AI Model Evaluation, Security Testing, Performance Testing)
- **Chapter 9** - Results and Discussion (Sections are Performance Metrics, Evaluation of Results. It must includes screenshots, analytics, AI results, workflow results)

- **Chapter 10** - Project Management and Workflow (Includes Git strategy, branch flow, collaboration workflow, development methodology, sprint/task management)
- **Chapter 11** - Conclusion and Future Work (It must mention optional features in the proposal and additionally distributed scaling, advanced analytics, and cloud native deployment)
- References

Exclusions

The following chapters/sections must be excluded totally:

- Complete code Listings
- Huge Datasets
- Excessive Theoretical Networking Concepts